



FIRST SEMESTER 2026-27
Course Handout (Part II)

Date: 13/08/2026

Course No : BITS U103
Course Title : Engineering Design & Prototyping
Instructor-in-charge : GEETA BHATT
Co-Instructors : AMIT KUMAR, SUNIL SINHMAR, GAURAV SHARMA
Practical Lab Instructors : Vikrant Pachouri, Anirudh Sharma, Durga M, Kunal Dhadse, Akshata Trivedi, Prafful Negi, Rajneesh Vishwakarma, Vedanshu Shaker Srivastav, Sachin Shishodia, Pronamika Borthakur, Shivendra Singh, Soni Kumari, Suryansh Lal, Adnan Tauqueer Farooqi, S. Sooraj Subramaniam, Amit Vashist, Raghav Agarwal, Rajnish Kumar Jha, K Nirmal Das, Vipul Maurya, Vignesh Ganesan, Karishma Chauhan, Devi Chandhana S

Scope and Objective of the Course:

Students will learn to put their design ideas in the form of drawings and further go on to create a prototype in the workshop. Engineering drawings and hands-on fabrication are blended into a foundational design course. It transitions students from design ideas to theoretical drawing techniques and finally to real-world prototyping, thereby imparting both software & shop-floor competencies.

Learning outcomes:

- Develop a foundational understanding of product design and prototyping.
- Engage in the practice of translating design concepts into technical drawings.
- Acquire practical experience in product and prototype development.
- Analyse the impact of design decisions on manufacturing feasibility & outcomes.

General Course Description:

Engineering design, Prototyping, Stages of new product development, Translating design ideas into 2D sketches, Theory of projection, Orthographic Projection drawings, Isometric drawings, Projection of lines, Projection of planes, Projection of solids, Section of solids, Development of surfaces, Characteristics of prototypes, Concept of fidelity, Importance of prototypes in new product development, Basics of traditional and modern manufacturing processes for prototyping, metrology for measurement, fits and tolerances, Design iteration for enhancing manufacturability, End-to-end design and development of prototypes (hands on exercises).

Textbook(s):

1. **TB1:** Madsen, D.A., Madsen, D.P., 2016. Engineering Drawing and Design, 6th Ed. Cengage Learning
2. **TB2:** Hallgrímsson, B., 2019. Prototyping and modelmaking for product design, 2nd Ed., Laurence King Publishing

TB1: Product design workflow and Drawings; **TB2:** Prototyping concepts and case studies.

Reference book(s)



1. **RB1:** Kulkarni, D. M., Rastogi, A. P., Sarkar, A. K., 2013. Engineering Graphics with AutoCAD, Revised Edition, PHI Publishing.
2. **RB2:** Dotson, C.L., 6th Ed., 2016. Fundamentals of Dimensional Metrology, Cengage Learning.
3. **RB3:** Ghosh, A., Mallik, A. K., Manufacturing Science, East-West Press Pvt. Ltd.
4. **RB4:** Chattopadhyay, A. B., Machining and Machine Tools, Wiley India Pvt. Ltd.

RB1: Engineering drawings and visualization concepts; **RB2:** Metrology-related concepts; **RB3 & RB4:** Manufacturing processes for prototyping.

Course Plan:

Module	Lecture/Lab Content	Reference	Learning Objectives
1. Introduction to EDP and related basics	L1: Introduction to EDP and Engineering Drawing L2: Product Design Thinking, Ideation Techniques and Sketching for Product Design	TB1: Ch1, Ch2, Ch5 RB1: Ch1	1. Understand the stages of the product lifecycle and practice critical product observation techniques 2. Apply design thinking and ideation strategies to conceptualize ideas through free-hand sketching
	Lab 1 (Practice and Eval): Product Observation and freehand drawing		
2. Theory of Projections	L3: Theory of Projections L4: Orthographic Projections L5: Orthographic Projections/Isometric projection	TB1 Ch3, Ch7, Ch8 RB1: Ch2, Ch3, Ch5	1. Perform basic operations in AutoCAD (2D) 2. Navigate the AutoCAD 2D interface to apply industry-standard dimensioning to product designs 3. Learn to draw the orthographic projections for given objects (pictorial views)
	Lab 2 (Practice): Introduction to AutoCAD-2D Lab 3 (Practice): Orthographic projections Lab 4 (Eval): Orthographic projections		
3. Isometric drawing and 3D part modelling	L6: Isometric projections L7: Introduction to AutoCAD 3D interface L8: 3D part modelling	TB1: Ch3, Ch14 RB1: Ch6	Learn to draw the isometric drawing with the given orthographic views and transition to the AutoCAD 3D interface for developing 3D part modelling skills
	Lab 5 (Practice): Isometric drawing Lab 6 (Eval): Isometric drawing Lab 7 (Practice): Introduction to AutoCAD-3D (Basic operations) Lab 8 (Practice): 3D part modelling Lab 9 (Eval): 3D part modelling		



4. Fundamentals of metrology and basic manufacturing processes/prototyping techniques	L9: Introduction to Metrology L10: Introduction to Fits and Tolerances L11: Fits and Tolerances L12: Fundamentals of Manufacturing processes mapped to CAD operations L13: Product fabrication planning L14: Rapid Prototyping and 3D printing L15: Material selection for 3D printing		
	Lab 10 (Demo + Practice): Dimensioning-Metrology Lab 11 (Demo + Practice): Sheet metal bending and laser cutting Lab 12 (Demo + Practice): Material Removal process mapped to CAD operations (lathe, shaper, milling, and drilling) Lab 13 (Demo + Practice): 3D printing + ironing Lab 14 (Demo + Practice): Grinding operation + welding	RB2: Ch1, Ch2, Ch3 RB3: Ch 1-6 TB1: Ch4, Ch10, Ch13 TB2: Ch1, Ch2, Ch9	1. Learn basic metrology techniques to accurately gather and interpret dimensional data of physical products 2. Apply principles of fits and tolerances to generate 3D parts using precise dimensions from measured components 3. Plan product fabrication workflows, map fundamental manufacturing processes to their corresponding CAD operations, select appropriate materials, and interface software for 3D printing
5. CAM and CNC	L16: Introduction to CAM and CNC L17: CAM and CNC		
	Lab 15 (Demo + Practice): CAM/CNC	RB4: Ch 25	Understand the basics of CAM and CNC machining
6. Projections of lines/planes/solids and sectioning	L18: Projections of Lines and Planes L19: Projections of solids L20: Section of solids		
	Lab 16 (Practice): Projection of lines/planes Lab 17 (Practice): Projection/Section of Solids Lab 18 (Eval): Projection/Section of Solids	RB1: Ch9, Ch10, Ch12, Ch13 TB1: Ch9, Ch12	Learn to draw the projection of lines/planes/solids and sectioning
7. Design review 1	L21: Case Study 1		
	Lab 19 (Eval): Design Review 1 - Concept, Drawing, partial CAD	TB1: Ch25	Conduct visual integration and design reviews



8. Development of surfaces	L22: Development of Surfaces and Sheet Metal Bending	RB1: Ch14 TB1: Ch19, Ch23	Learn the principles of surface development and sheet metal bending as observed in practical manufacturing demonstration
	Lab 20 (Practice): Development of surfaces Lab 21 (Eval): Development of surfaces		
9. Prototype Fidelity and Design Iteration	L23: Prototype development, Prototype Fidelity and Design Iteration	TB1: Ch25, Ch3, Ch4 TB2: Ch2	1. Work with several machines to develop a prototype 2. Evaluate prototype fidelity for design iteration and observe finishing processes, including lathe, drilling, and 3D printing 3. Design user manuals and construct low-fidelity prototypes for hands-on practice
	Lab 22: Prototype development in workshop Lab 23: Prototype development in workshop Lab 24: Prototype development in workshop + Discussion on Low Fidelity Prototype Design		
10. Product finalization	L24: Product Documentation L25: User Manual Design L26: Final Report Formatting and Design Expectations	TB1: Ch15, Ch25	Format the final report to meet design expectations and undergo the second design review to test product workability and suggest necessary iterations
	Lab 25 (Eval): Design Review 2 - Testing, Product workability, suggest iterations		

Evaluation Scheme (Total Marks: 300):

Modules	Marks	Dates/Venue	Percentage
Attendance-associated discussion-Workshop	18	Workshop (all non-evaluative sessions)	6 %
Lab Assignments (Evaluatives) (Best 4 out of 6)-No makeup	70	As per the dates announced by IC in the Lab (workshop/IPC)	23.33 %
Project	52	Two design review meetings	17.33 %
Mid-semester test (On prototyping)	60	As announced by AUGSD	20 %



Comprehensive exam	100	- Written exam details will be communicated. - Computer test will be on date announced by AUGSD as per the schedule sent by IC.	33.33 %
1. Written exam (on prototyping)	(40)		
2. Computer-based test (on Design portion)	(60)		
Course Total	300		

Consultation Hours:

1. Prof. Geeta Bhatt: Thursday, 11:00 AM-12:00 PM, 6020-C.
2. Prof. Amit Kumar: Tuesday, 10.30 AM-11:30 AM, 2220-J.
3. Prof. Sunil Sinhar: Monday, 12:00 PM- 01:00 PM, 2220-L.
4. Prof. Gaurav Sharma, Thursday, 11:00 AM-12:00 PM, 6020-J.

Notices: All notices and practice videos will be sent via email/google space group “EDP Announcements_26-27_Sem I”.

NC criteria:

The passing marks for this course will be determined based on one of the two criteria: 30% of the median of the total course score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as passing marks in this course.

Attendance Policy:

- Each student is expected to attend all the classes and participate in all the evaluation components.

Make - up policy:

No make-up exams will be conducted for lab-based evaluation. Make-up requests for tests (Before mid-semester/ comprehensive examination) must be submitted at least one day before the exam date on the AUGSD portal only.

**Instructor-in-charge
(BITS U103)**